

Phase 03 - DP-DBBA Evaporation

Explanation and Operating Logic - Unified GUI

Release: 0.9.36 **Build:** 2026.08.11-r15

Operator and developer explanation of the current Phase 03 unified GUI, ratio handoff, evaporation logic, and distinct Abort versus Finish behavior.

1. Role in the synthesis workflow

- Use the Phase 01 thickness ratio to compute the CK-1 DP-DBBA target.
- Set the external oven PID target to 200 C and require independent oven stability before shutter readiness.
- Heat the CK-1 evaporator, guide shutter opening, track relative CK-1 thickness, then hand off to Phase 04.

2. Ratio confirmation before launch

Before every Phase 03 start, the unified launcher shows the stored Phase 01 ratio. Yes reuses it; No opens an editable value. If no session ratio exists, the operator enters a positive ratio manually.

Unified launcher: START_NPG_CHAMBER.bat opens the main launcher. Each phase starts as a separate child process and owns its serial ports and hardware cleanup.

Phase-local shutdown: Abort / Safe Stop closes only the active phase after its safe hardware action. The unified launcher remains open and can be used again after the child process releases its ports.

Launcher close protection: while a phase is active, the launcher itself refuses to force-kill it. Use the phase GUI Abort / Safe Stop or let the phase finish normally.

3. Heating and shutter transition

Condition	Requirement
External oven	Stable around 200 C inside the configured tolerance for the configured stability time.
CK-1 temperature	At or above the active target/guide.
Average CK-1 rate	At or above the requested rate target.
Controller-only diagnostics	Do not gate shutter opening.

The upper control-rate band is not a process gate. A rate above target does not prevent WAIT_SHUTTER_OPEN.

4. Evaporation and normal handoff

- Open Shutter confirmation resets the evaporation measurement window.
- The finish criterion is CK-1 relative thickness reaching the ratio-derived DP-DBBA target.
- At target, physically close the shutter and confirm it with Close shutter or Finish phase.
- Normal completion calls the Phase 04 handoff: set Keysight current to the base working current, default 0.640 A, and keep OUTPUT ON.
- Phase 03 then saves/cleans up and closes only its own process. The unified launcher remains open for Phase 04.

5. Abort versus Finish Phase

Action	Keysight	Oven PID	Result
Abort / safe stop	Immediately CURR 0.000 + OUTP OFF.	After Keysight is already OFF, request 0 C best-effort.	Close Phase 03 only; no slow abort ramp-down.
Finish phase	Return to 0.640 A base current and keep OUTPUT ON.	Keep normal Phase 03 process state.	Normal handoff, ready to start Phase 04.

Important: Finish phase remains blocked until WAIT_SHUTTER_CLOSE. This prevents a normal handoff before the deposition target has been reached and the operator is at the shutter-close step.

Abort ordering: Keysight power removal occurs before the PID 0 C transaction. A slow or failed PID response can never delay OUTP OFF.

6. Cleaned configuration

Phase 03 no longer exposes obsolete abort-ramp-down step/period parameters. They are not used by either valid Phase 03 ending: Abort is immediate OFF, while normal Finish is the 0.640 A / OUTPUT ON handoff.

7. Safety and data

- Automatic current cap remains 0.660 A; software hard stop remains 0.680 A; watchdog maximum remains 255 C by default.
- Pressure, QMB plausibility, Keysight protection and temperature watchdog logic remain active.
- Run outputs are stored under Data Samples/DP-DBBA Evaporation Data/.
- After Finish, verify the physical shutter is closed and the Keysight is at the expected handoff state before starting Phase 04.

Operational note: Software state is not a substitute for checking the real chamber hardware. Verify shutters, pressure, PID, Keysight output and any manually operated hardware before continuing.